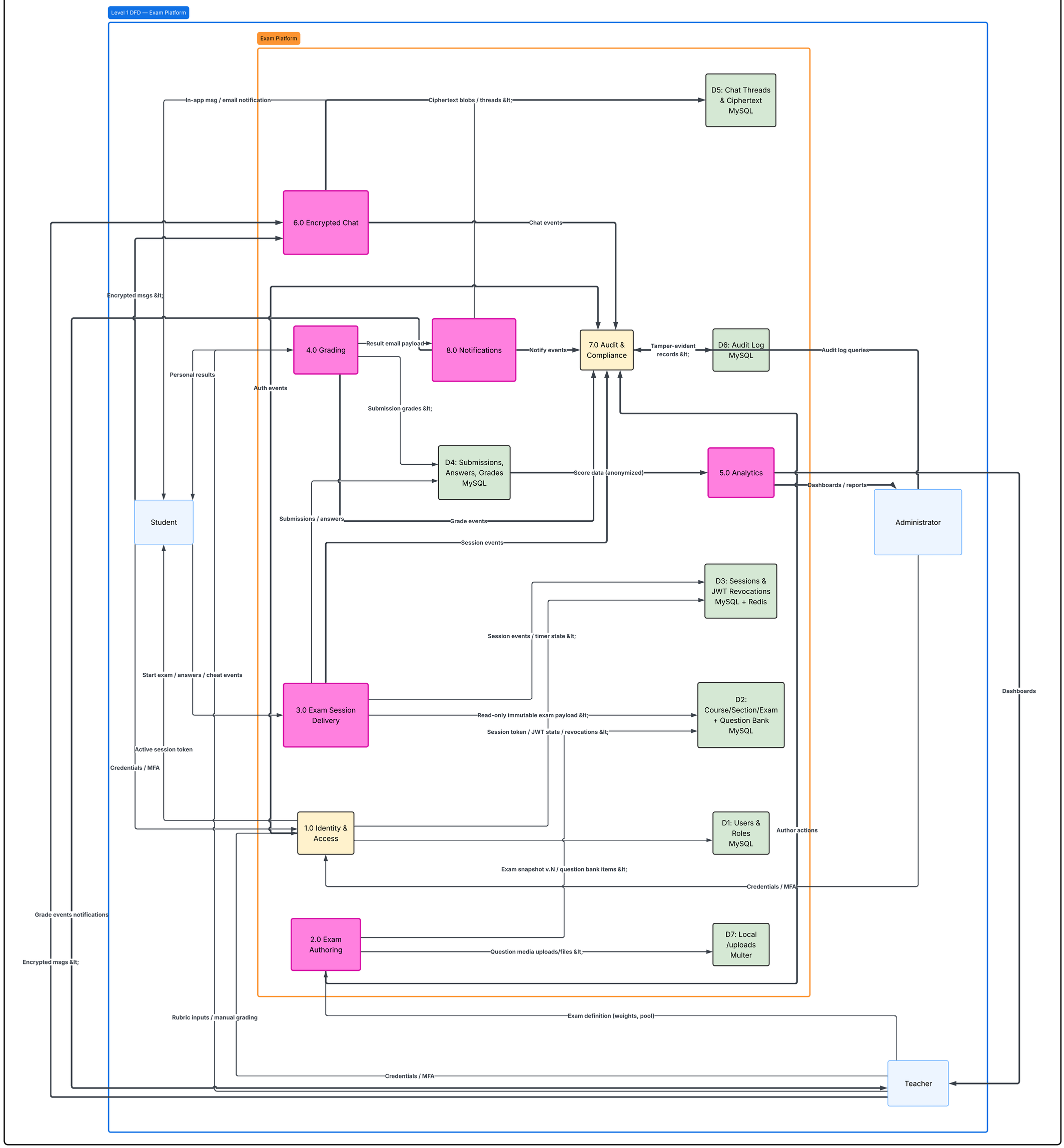
6. DFD Level 1


Level 1 DFD — Exam Platform
Exam Platform
D5: Chat Threads & Ciphertext MySQL
In-app msg / email notification
Ciphertext blobs / threads <
6.0 Encrypted Chat
Chat events
Encrypted msgs <
4.0 Grading
Result email payload
8.0 Notifications
Notify events
7.0 Audit & Compliance
Tamper-evident records <
D6: Audit Log MySQL
Audit log queries
Personal results
Auth events
Submission grades <
D4: Submissions, Answers, Grades MySQL
Score data (anonymized)
5.0 Analytics
Dashboards / reports
Student
Submissions / answers
Grade events
Session events
Administrator
D3: Sessions & JWT Revocations MySQL + Redis
Session events / timer state <
Start exam / answers / cheat events
3.0 Exam Session Delivery
Read-only immutable exam payload <
D2: Course/Section/Exam + Question Bank MySQL
Session token / JWT state / revocations <
Active session token
Credentials / MFA
1.0 Identity & Access
D1: Users & Roles MySQL
Author actions
Exam snapshot v.N / question bank items <
Credentials / MFA
Dashboards
Grade events notifications
2.0 Exam Authoring
D7: Local /uploads Multer
Question media uploads/files <
Encrypted msgs <
Rubric inputs / manual grading
Exam definition (weights, pool)
Credentials / MFA
Teacher